Sean D. White

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EDUCATION-

Merrimack College, North Andover, MA — May 2025

Bachelor of Science in Mechanical Engineering

Honors: Graduated **Summa Cum Laude**, GPA: 3.9**,** President's List Recipient

SKILLS-

- **Software: CAD Certified**, **SolidWorks**, Autodesk Inventor/AutoCAD, MATLAB, Oscilloscope, Arduino, Adobe Illustrator, Adobe After Effects, Adobe Photoshop, Microsoft Word, PowerPoint, Laser Marking software, Excel, and Publisher
- **Equipment: Machine Shop certified**, Use of a variety of shop power saws (radial arm saw, band saw, table saw), Grinder, Dremel, Abrasive Blasting, Drill Press, Various sanders, General Hand Tools, 3D Printers (FDM, SLA)
- **Personal:** Measurement Taking, Data analysis/retrieval, Mechanical Design, Analytical and Critical Thinking, Failure Prevention, Advanced Mathematical Abilities, Teamwork, Communication, Writing

ENGINEERING POSITIONS-

ARCH Medical Solutions– *Laser Applications Engineer,* Seabrook, NH — Jan 2026- Present

- Lead Program Development: Served as the primary laser program builder for the facility, specializing in FOBA laser systems while managing Beamer, TRUMPF, and Rofin hardware/software workflows.
- Engineering Design and Tooling: Utilize SolidWorks to design/3D print custom fixtures for flatbed and rotary marking, optimizing part alignment and repeatability. Also developed complex DXF files.
- GD&T understanding: Analyzed and interpreted Engineering drawings regularly.
- Developed expertise in laser-material interaction by systematically manipulating power, frequency and scan speed to achieve precise, durable, and legible marking results across diverse materials.
- Perform rigorous Quality Control verification using Keyence measurement systems, barcode scanning (traceability) to ensure 100% adherence to engineering specifications
- Lean Implementation: Led the transition to a 5S environment to reduce setup errors and increase overall workflow efficiency.

MK Servies Corp – *Quality Control Intern/ 6S Technician,* Middleton, MA — May 2024- August 2024

- Implemented a 6S system for the entire facility (milling, shipping, lathe, swiss)– Improved efficiency, reduced waste, and led to a more organized, clean workspace.
- Shadowed machinist and received introductory training on CNC machining.
- Worked with advanced QC instruments – a manual CMM, an Oasis Inspection system, Micro-Hyte, and a comparator.
- Became proficient at using micrometer and calipers to inspect critical parts and write inspection reports.

CADSPARC LLC - *CAD Design Intern***,** Haverhill MA — May 2022 – August 2022

- Design consultancy helped take products from initial concepts to physical prototypes.
- Utilized industrial grade 3D printers (FDM, SLS).
- Focused on CAD modeling and fast prototyping to improve customer designs.

PROJECTS (Skills Attained)-

Senior Design Project – *Robotics Competition Design* — Spring 2025

- Collaborated in diverse teams integrating multiple disciplines (ME, EE, CS), delivering a fully autonomous VEX robot.
- Designed a pneumatic climbing mechanism, optimizing geometry and actuator placement for mechanical advantage; validated through FEA at critical load stages achieving repeatable lifting performance.
- Conducted comprehensive stress analysis determining failure sequence of the climbing arm assembly.
- Applied fatigue analysis to verify an appropriate lifespan under competition loading conditions.

Advanced Mechanics, *Lancia Fulvia V4 Engine Build/Analysis* — Fall 2024

- Modeled complete Lancia Fulvia V4 Engine assembly in Autodesk Inventor, including crankshaft, pistons, connecting rods, and flywheel components.
- Used Thermodynamics to calculate peak cylinder pressure and combustion forces
- Conducted finite element analysis on connecting rods under combustion loading to determine stress distribution and displacement values (validated FEA with hand calculations).
- Executed dynamic simulation to analyze stress variations during combustion.

Experimental Mechanics, *Digital Image Correlation* — Spring 2025

- Performed non-contact optical measurement techniques to analyze full-field displacement and strain patterns in materials under tensile loading using MATLAB- based Ncorr software.
- Identified and quantified stress concentrations around geometric discontinuities/material heterogeneities.

Heat and Mass Transfer, *Heat Exchanger Design and Construct* — Spring 2024

- Designed a built multi- pass shell and tube heat exchanger within 18" length constraint.
- Used heat transfer principles to validate performance and optimize design.